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NAAC GRADE A+
ACCREDITED UNIVERSITY

Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 4

Title: Data Cleaning and Transformation of Cafe Sales Dataset using Power BI (Power Query)

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Date: 11/09/2026

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Experiment No.: 4

Data Cleaning and Transformation of Cafe Sales Dataset using Power BI (Power Query)

GitHub-Repo-Link: prakruthidevanga/Business-Analytics: Business Analytics

1. Problem Statement

I was working as a Data Analyst with customer transaction data from a cafe. The raw sales data, received as a CSV file, contained missing values, duplicate records, and inconsistent entries such as placeholder text in place of real values. Before this data could be used for business reporting or analysis, it had to be imported into Power BI and cleaned using Power Query so that it became accurate, consistent, and ready for downstream use.

2. Methodology / Steps Followed

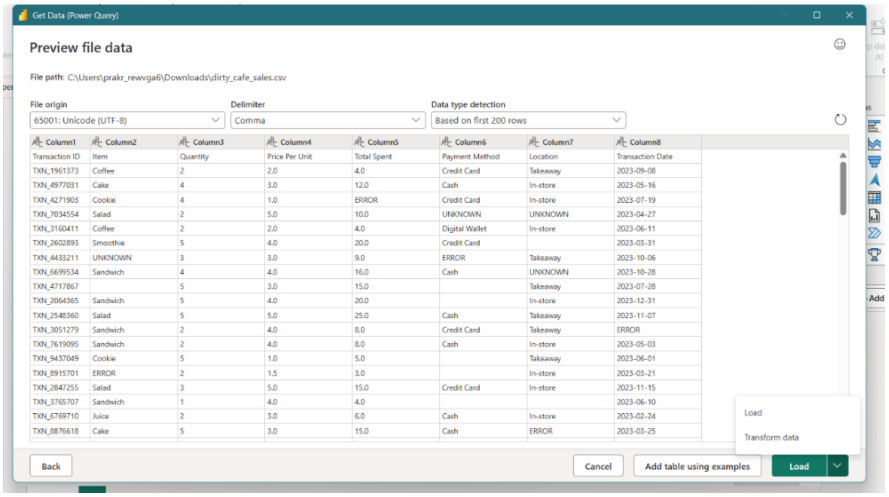
Step 1: Data Collection

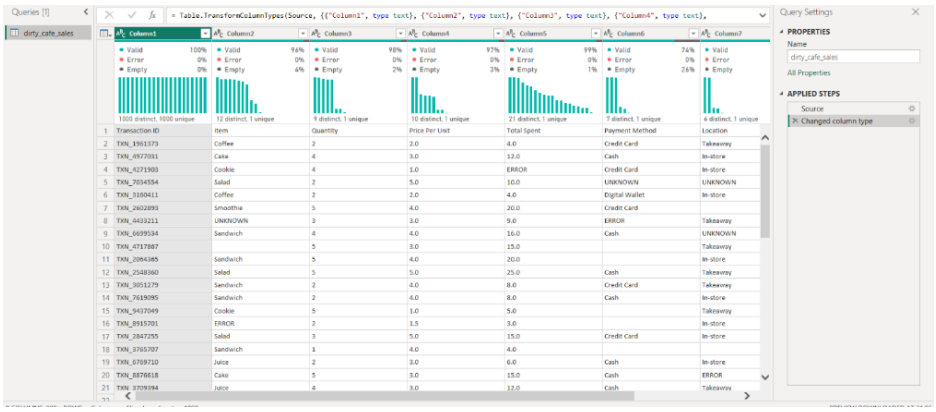
I obtained the Cafe Sales dataset (dirty_cafe_sales.csv) containing 10,000 transaction records across 8 columns: Transaction ID, Item, Quantity, Price Per Unit, Total Spent, Payment Method, Location, and Transaction Date. The dataset was sourced in raw form for the purpose of practicing data cleaning, and was known to contain missing values, duplicate records, and inconsistent placeholder entries.

Dataset Link: https://www.kaggle.com/datasets/ahmedmohamed2003/cafe-sales-dirty-data-for-cleaning-training

Step 2: Data Import

- Opened Power BI Desktop.
- Used Home → Get Data → Text/CSV to import dirty_cafe_sales.csv.
- Selected Transform Data to open the Power Query Editor instead of loading the raw file directly, so that the data could be cleaned before it reached the report.
- All 8 columns were initially loaded and set to Text type, since the file contained mixed placeholder values that prevented automatic type detection letting Power BI auto-detect types at this stage would have caused type-conversion errors on numeric columns like Quantity and Total Spent.

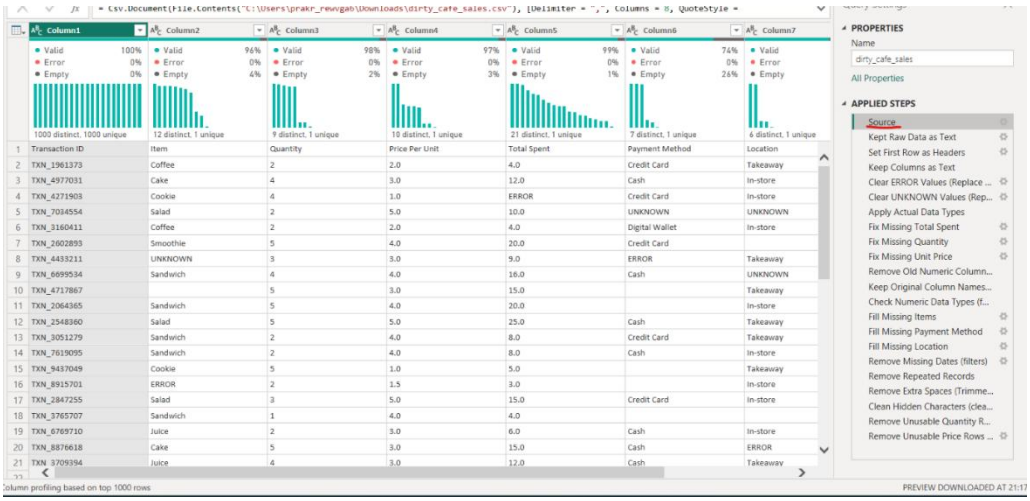




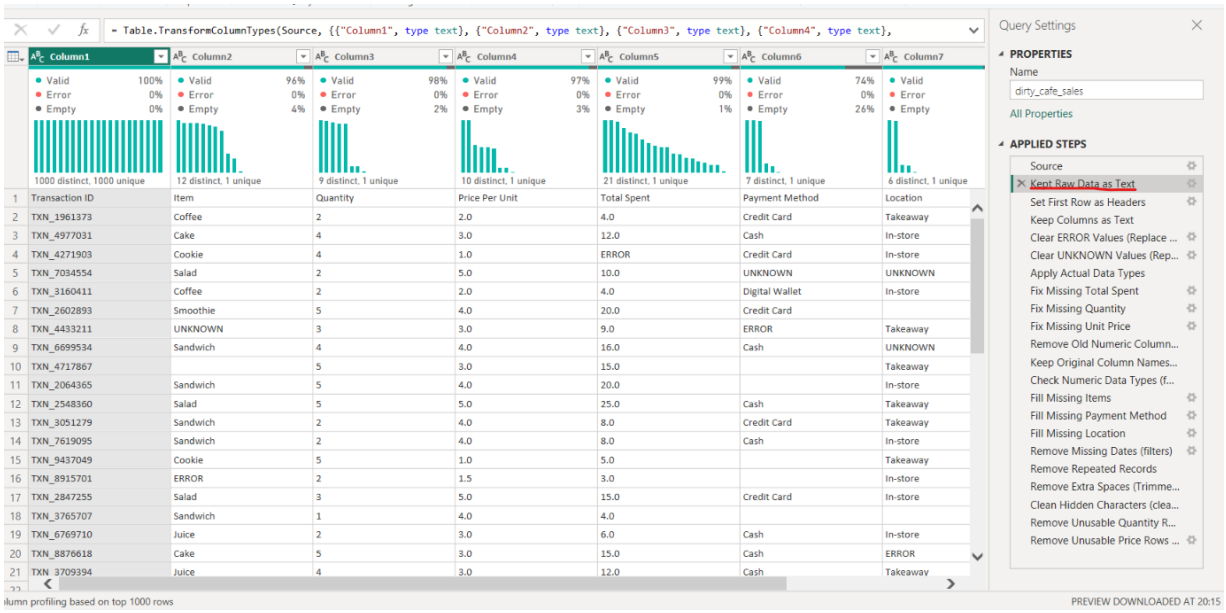
Step 3: Data Cleaning and Transformation (Performed in Power Query)

The following transformations were applied inside the Power Query Editor, in order. Each step appears in the Applied Steps pane under its default Power Query name; I renamed each one to something clearer, so the process reads more easily on review.

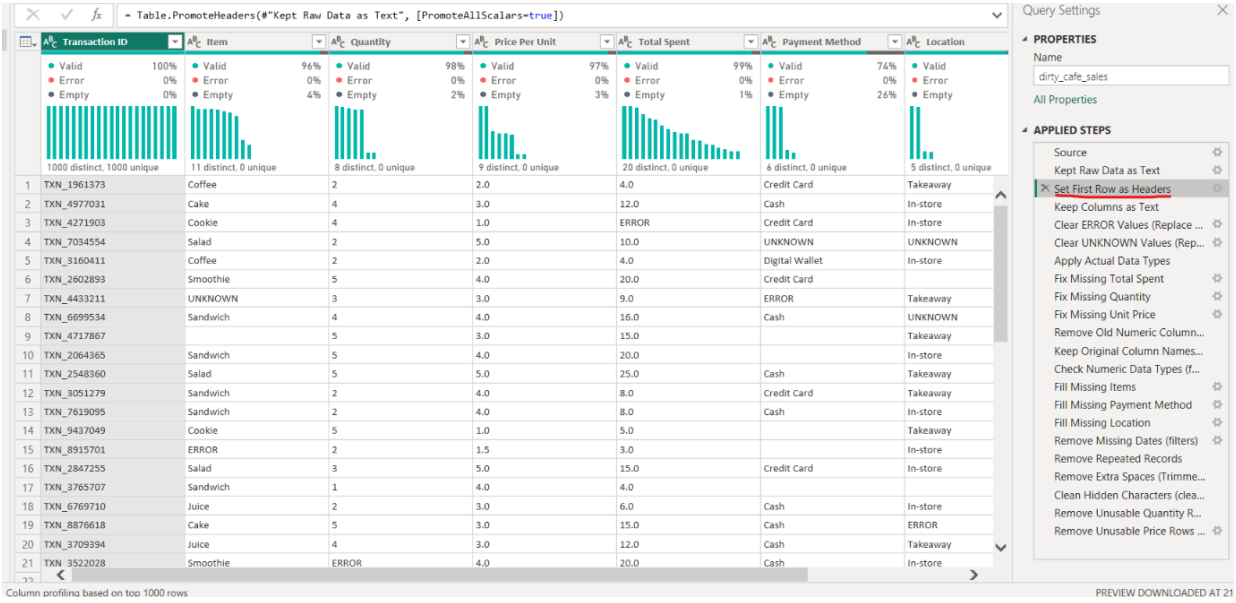
1. **Source keep as "Source":**
    - Connects to the CSV file and loads the raw data.
    - This step forms the base of the whole query — every later step builds on top of it, so I never modified it directly.



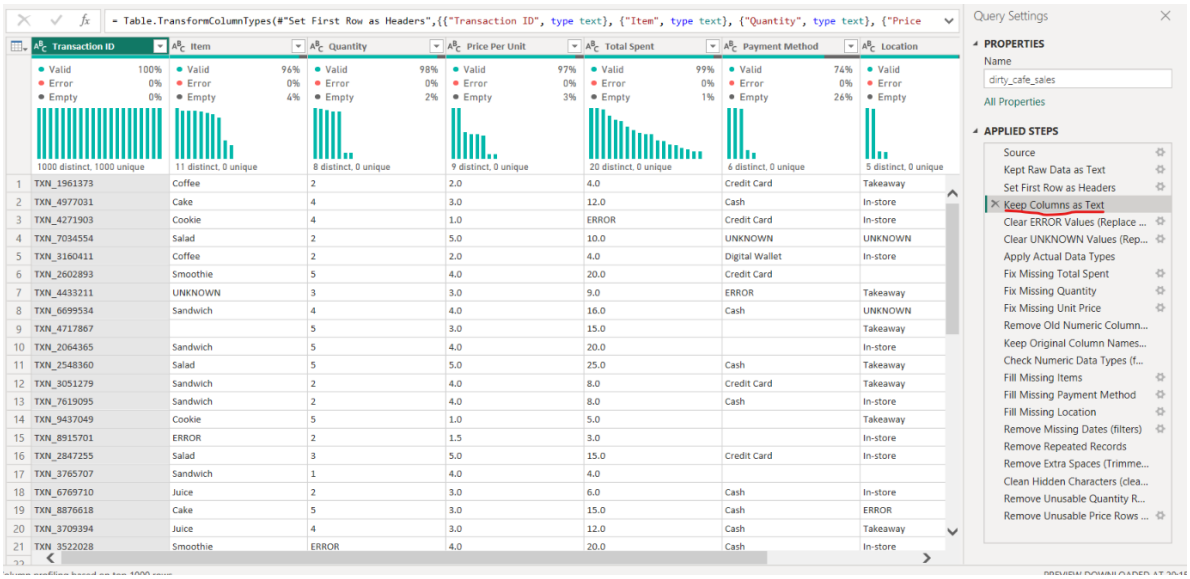
2. **Changed Column Type → rename to "Kept Raw Data as Text":**
    - Converted all columns to Text immediately after import, since raw values were mixed with placeholder text.
    - I did this deliberately as a safety step, because if any column had been left at its auto-detected type, Power Query might have tried to convert "ERROR"/"UNKNOWN" text into numbers or dates and thrown errors before I even had a chance to clean them.



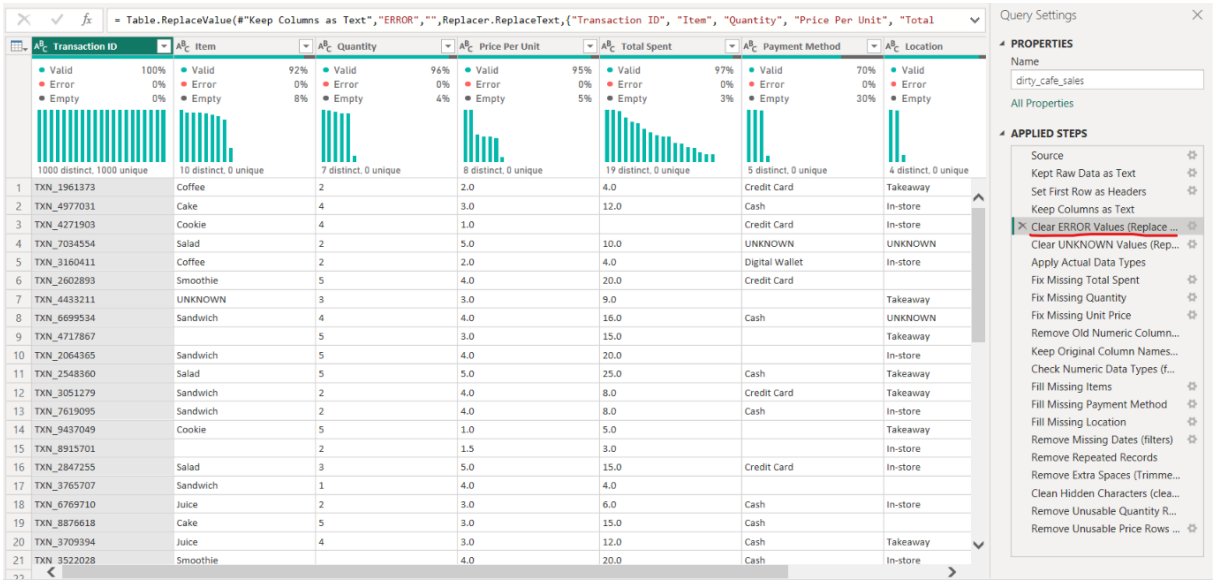
3. **Promoted Headers →rename to "Used First Row as Headers":**
    - Promoted the first row of the CSV to become the actual column headers (Transaction ID, Item, Quantity, etc.).
    - Without this step, the header row would have been treated as a normal data row, and my columns would have shown up as "Column1," "Column2," and so on instead of meaningful names.



4. **Changed Type → rename to "Keep Columns as Text":**
    - Re-applied Text type to all 8 named columns after the headers were promoted.
    - Promoting the headers can sometimes reset Power Query's type detection, so I reapplied Text type here just to make sure every column stayed locked as Text before I started removing placeholder values.



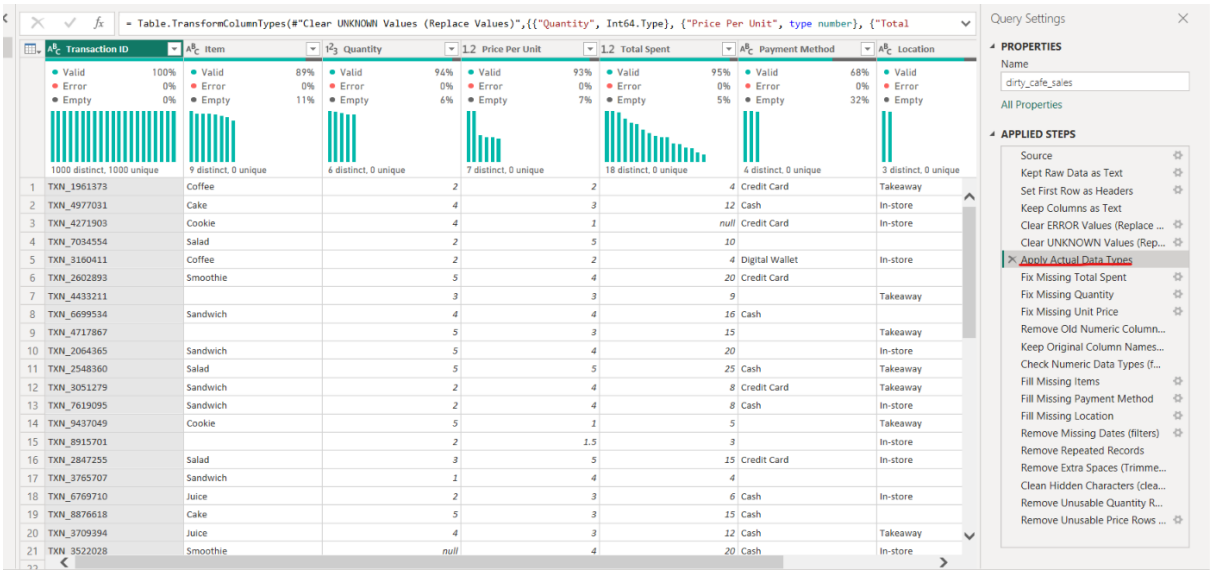
5. **Replaced Value → rename to "Clear ERROR Values (Replace Values)":**
    - Replaced every occurrence of the placeholder text "ERROR" across all columns with an empty value.
    - I applied this replacement across the entire table at once rather than column by column, since "ERROR" was appearing inconsistently across different columns and doing it column-by-column would have taken much longer.



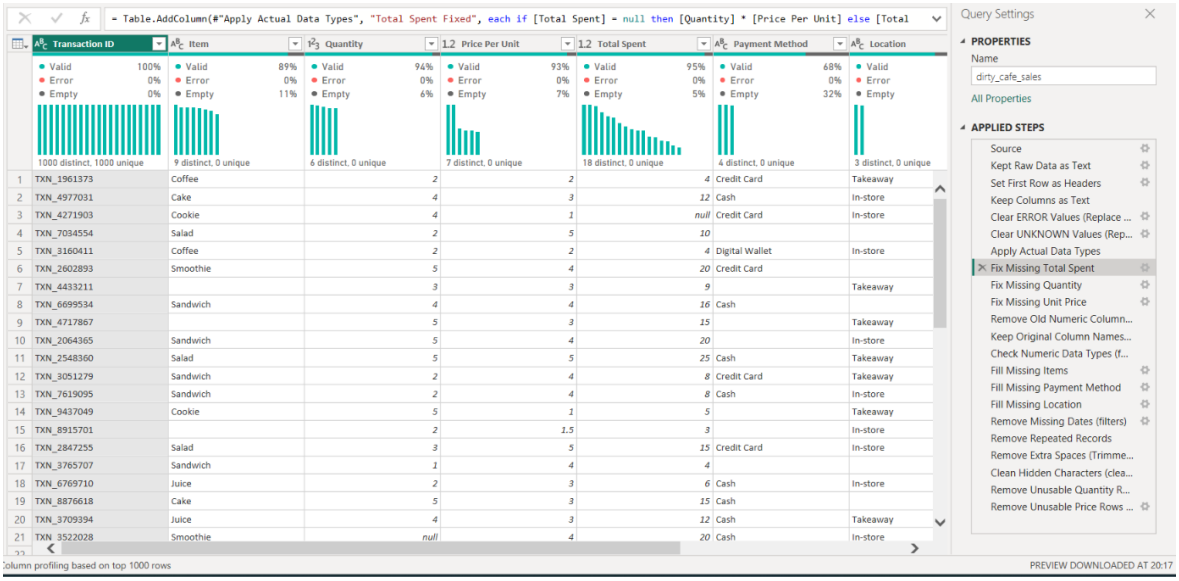
6. **Replaced Value1 → rename to "Clear UNKNOWN Values (Replace Values)":**
    - Replaced every occurrence of the placeholder text "UNKNOWN" across all columns with an empty value.
    - I kept this as a separate step from the "ERROR" replacement rather than combining them, since keeping them separate made it easier to check how many values each individual placeholder was affecting, and to isolate any mistakes.



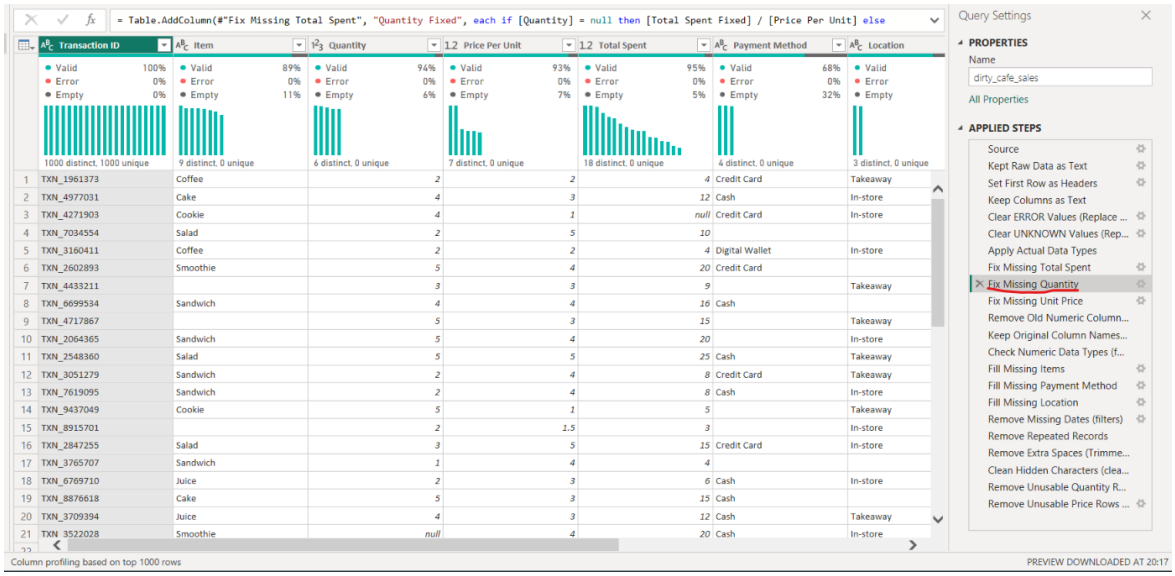
7. **Changed Type1 → rename to "Apply Actual Data Types":**
    - Converted Quantity to Whole Number, Price Per Unit and Total Spent to Decimal Number, and Transaction Date to Date.
    - This step could only run cleanly because the ERROR/UNKNOWN placeholders had already been removed in Steps 5 and 6 — if I had tried converting types before removing them, the conversion would have failed on almost every row, so the order of these steps genuinely mattered.



8. **Added Custom → rename to "Fix Missing Total Spent":**
    - Added a calculated column that fills a blank Total Spent using Quantity multiplied by Price Per Unit.
    - I used an if...then...else condition here — if Total Spent was null, it calculated Quantity × Price Per Unit; otherwise, it kept the existing value untouched.

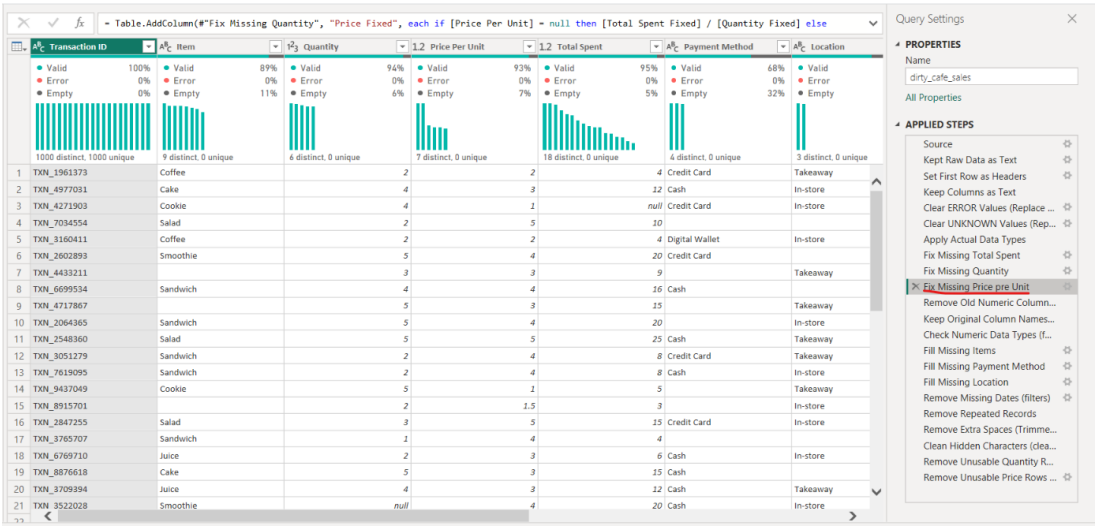


9. **Added Custom1 → rename to "Fix Missing Quantity":**
    - Added a calculated column that fills a blank Quantity using the (newly fixed) Total Spent divided by Price Per Unit.
    - This step depends on Step 8 already being complete, since it uses the newly recovered Total Spent value — referencing the original (still-blank) Total Spent column here instead would have produced incorrect results.



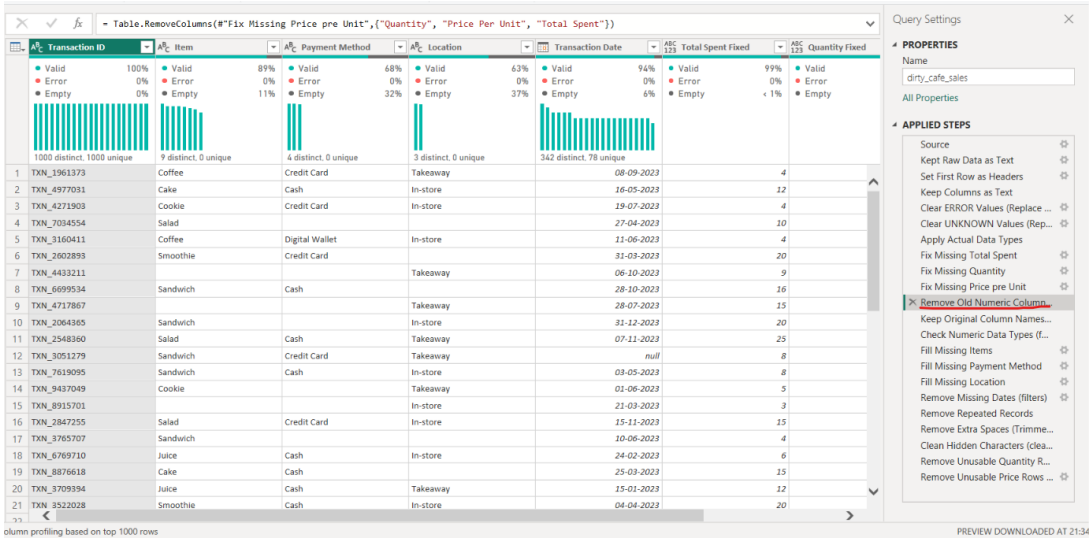
10. Added Custom2 → rename to "Fix Missing Price pre Unit":

- Added a calculated column that fills a blank Price Per Unit using the fixed Total Spent divided by the fixed Quantity.
- By this point, both Total Spent and Quantity were already reliable, so this final calculated column could safely use both together.

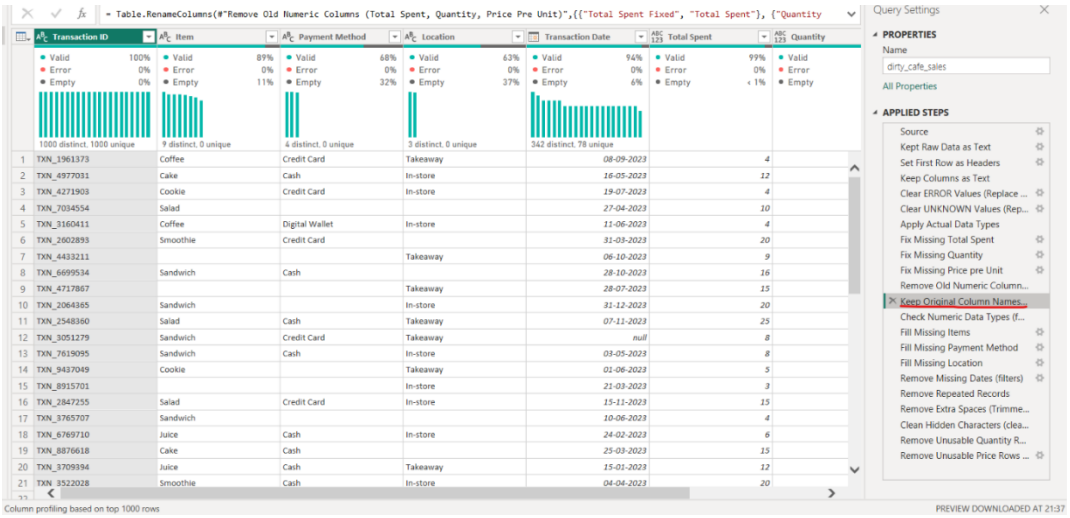


11. Removed Columns → rename to " Remove Old Numeric Columns (Total Spent, Quantity, Price Pre Unit) ":

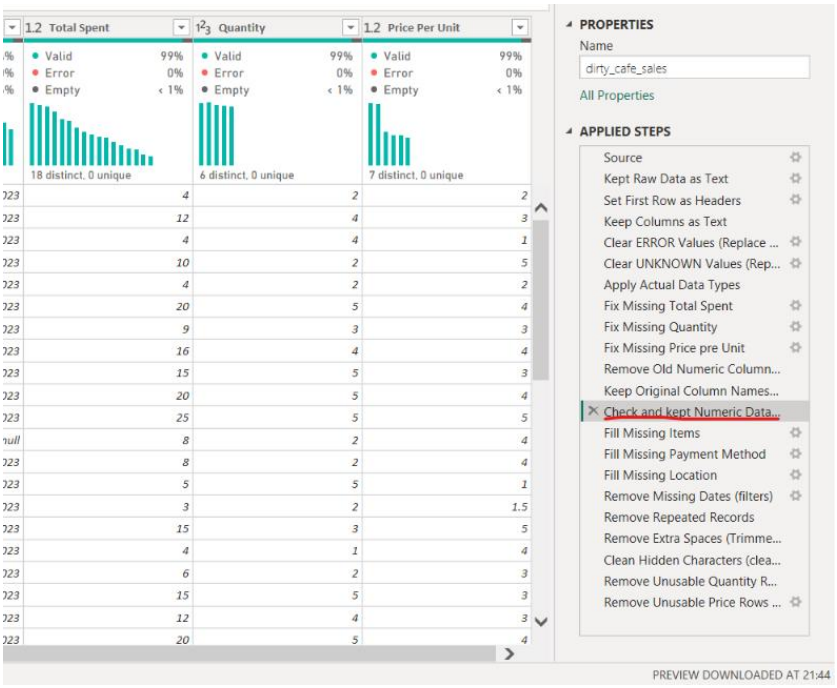
- Removed the original Quantity, Price Per Unit, and Total Spent columns, keeping only the newly calculated versions.
- At this stage the table temporarily had both the old (incomplete) and new (calculated) versions of these three columns side by side, so this step cleaned that up by dropping the outdated ones.



12. **Renamed Columns → rename to " Keep Original Column Names (for New columns named as original column name) ":**
    - Renamed the three calculated columns back to Quantity, Price Per Unit, and Total Spent.
    - I did this so that all later steps, and any future report visuals built on this table, could keep referencing the original column names without needing any changes.

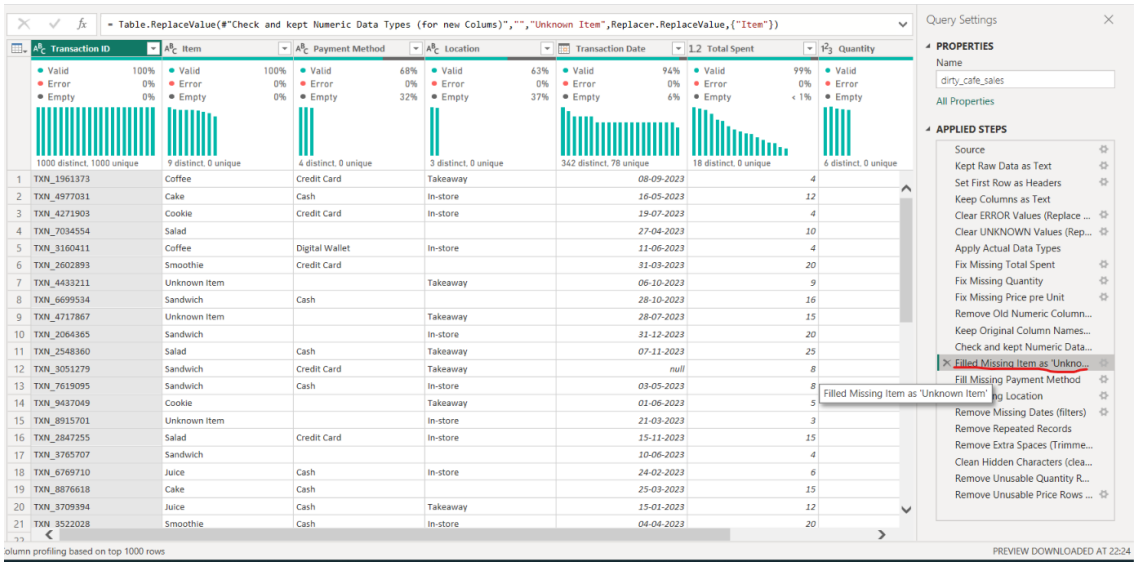


13. **Changed Type2 → rename to " Check and kept Numeric Data Types (for new Columns)":**
    - Reapplied Whole Number and Decimal Number types to the renamed columns (Total Spent and Price Pre Unit: Decimal Number, Quantity: Whole Number).
    - Newly added custom columns in Power Query default to a generic "Any" type, so this step was necessary to lock them back into proper numeric types.



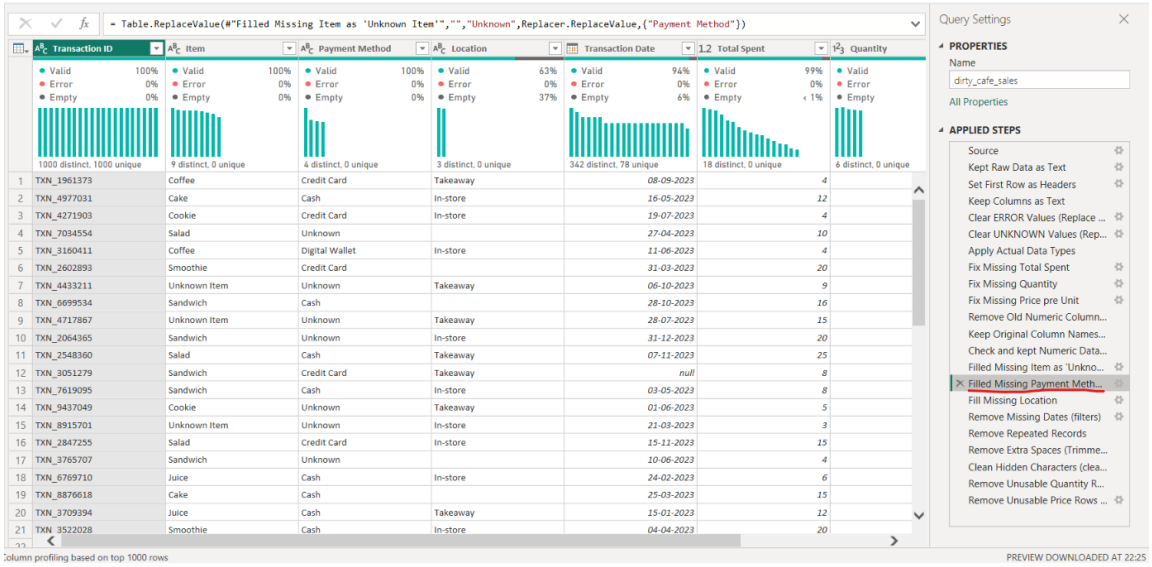
14. Replaced Value2 → rename to "Filled Missing Item as 'Unknown Item'":

- Replaced remaining blank values in the Item column with the label "Unknown Item".
- I chose to label this rather than delete the row, since a missing Item name didn't mean the rest of the transaction (Quantity, Price, Date) was unusable.



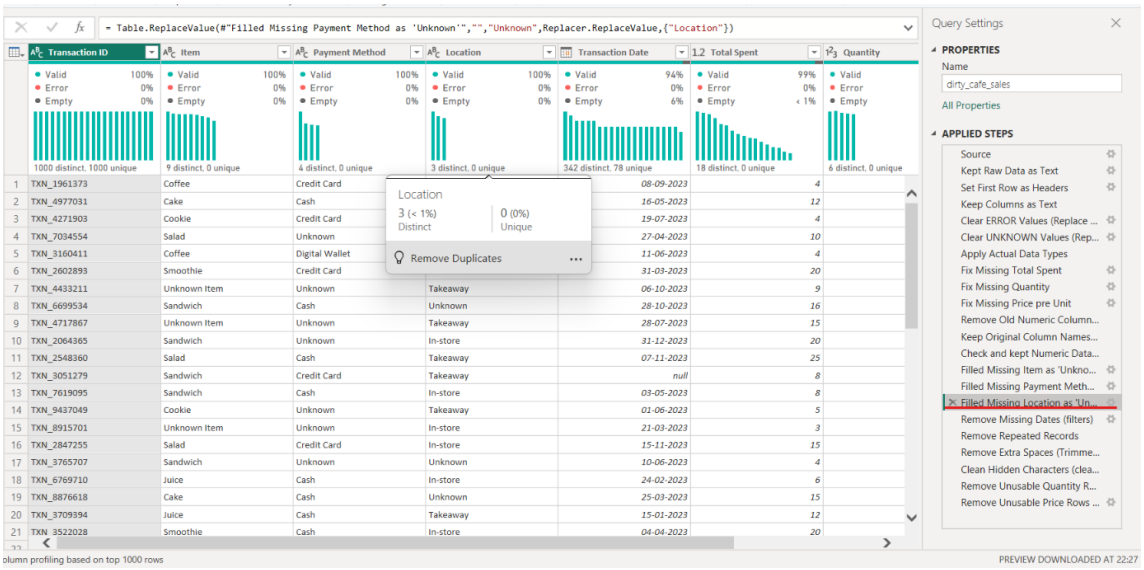
15. Replaced Value3 → rename to "Filled Missing Payment Method as 'Unknown'":

- Replaced remaining blank values in the Payment Method column with the label "Unknown".
- This preserved rows where only the payment method was missing, instead of losing an otherwise complete and valid transaction record.



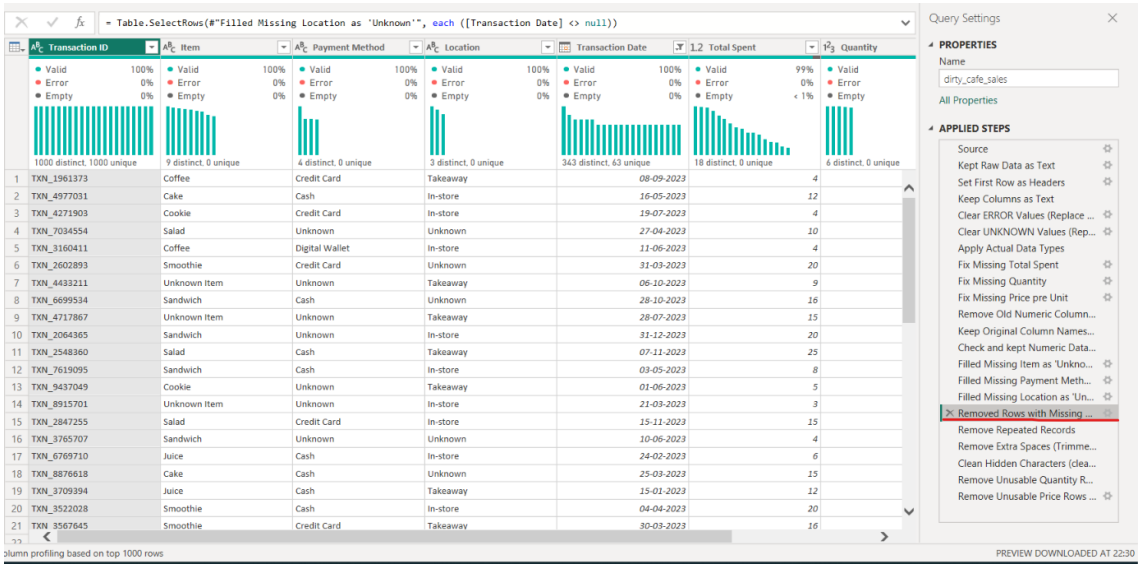
16. Replaced Value4 → rename to "Filled Missing Location as 'Unknown'":

- Replaced remaining blank values in the Location column with the label "Unknown".
- Since all three of these columns (Item, Payment Method, Location) had a fairly high proportion of missing values, deleting every row with a blank in any one of them would have discarded a large share of the dataset labelling them "Unknown" kept that data usable.



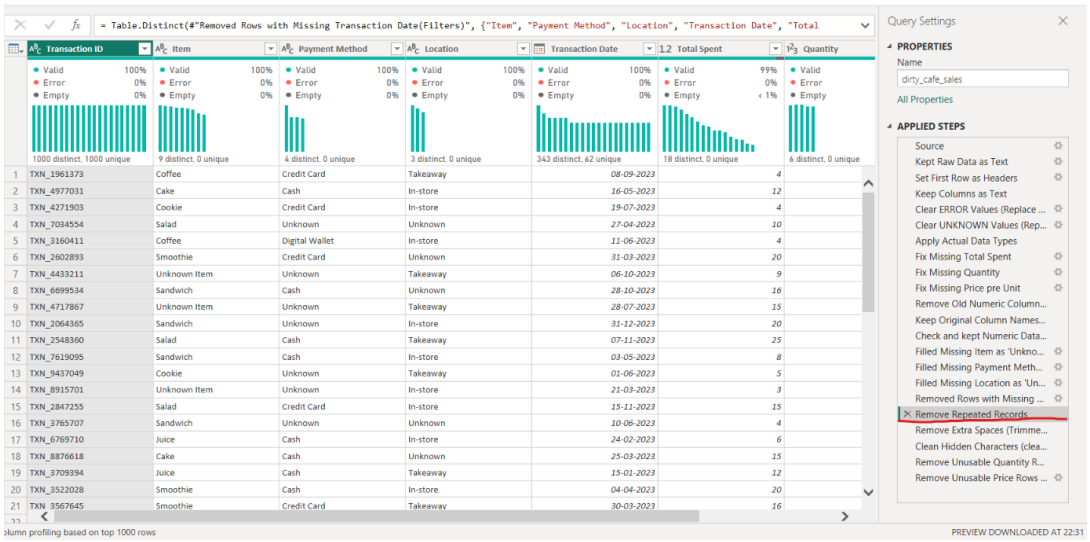
17. Filtered Rows → rename to "Removed Rows with Missing Transaction Date":

- Removed all rows where Transaction Date was blank, since a missing date could not be inferred from any other column.
- Unlike the numeric columns, there was no related field I could use to calculate or estimate a missing date, so removing these rows was the only reliable option.



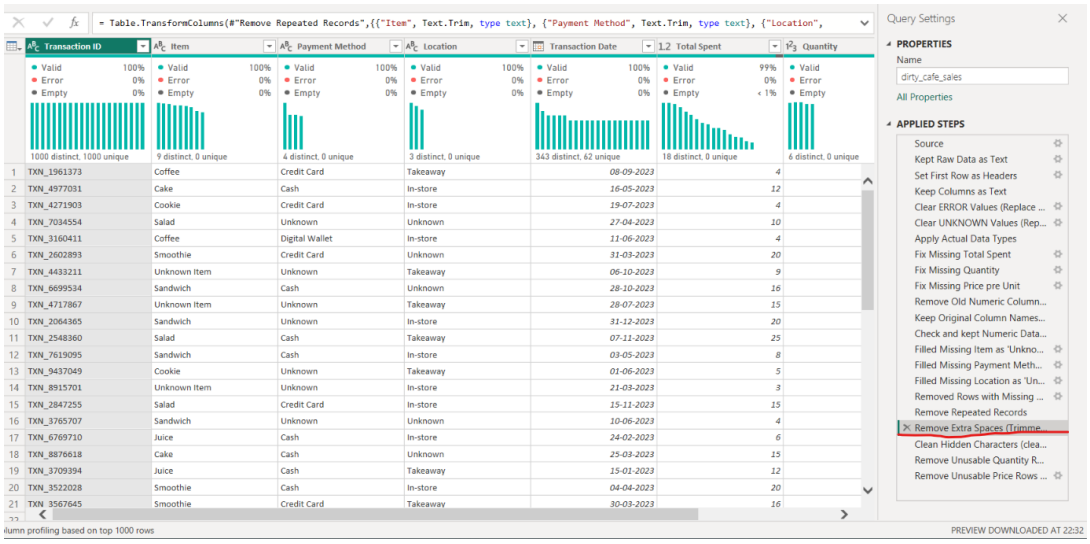
18. Removed Duplicates → rename to " Remove Repeated Records ":

- Removed rows that were identical across Item, Payment Method, Location, Transaction Date, Total Spent, Quantity, and Price Per Unit, keeping only the first occurrence of each.
- I deliberately excluded Transaction ID from this duplicate check — checking duplicates based on the actual transaction details is a more reliable way to catch true duplicates, in case the ID field itself was ever generated incorrectly.



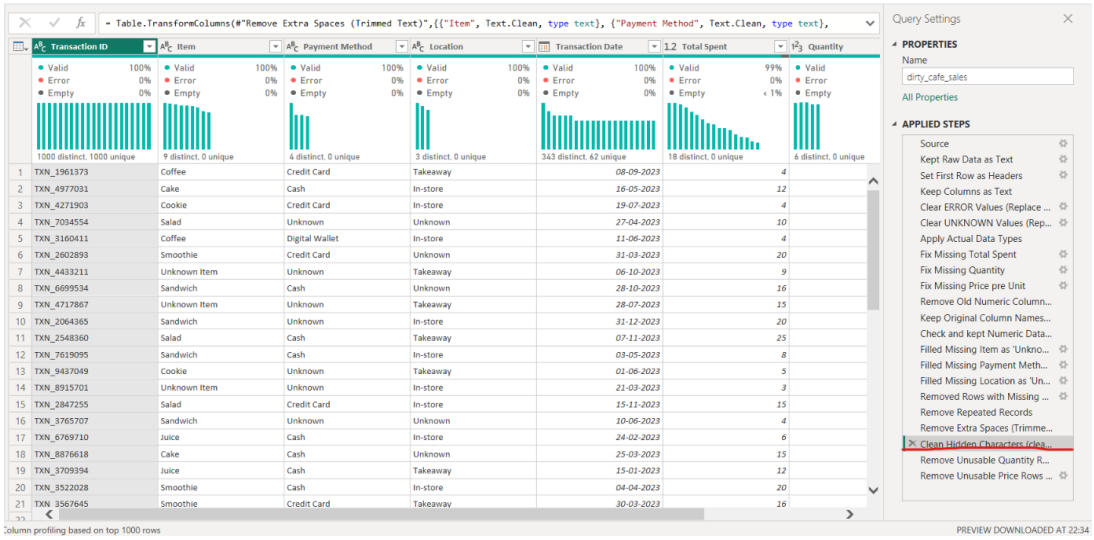
19. Trimmed Text → rename to " Remove Extra Spaces (Trimmed Text)":

- Removed leading and trailing spaces from the Item, Payment Method, and Location columns.
- This mattered because two values that look identical on screen, like "Coffee" and "Coffee " with a trailing space, would otherwise be treated as two separate categories by Power BI, silently breaking any grouping done later.



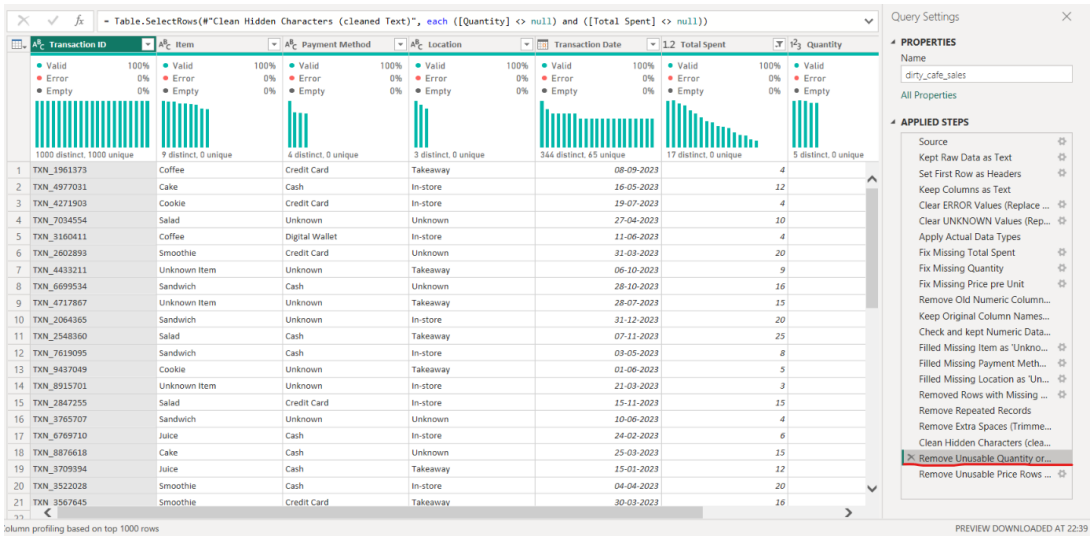
20. Cleaned Text → rename to "Clean Hidden Characters (cleaned Text) ":

- Removed hidden or non-printable characters from the same three text columns.
- These characters are invisible on screen but can still be introduced during CSV exports, and they can affect text matching and grouping just as much as extra spaces do.



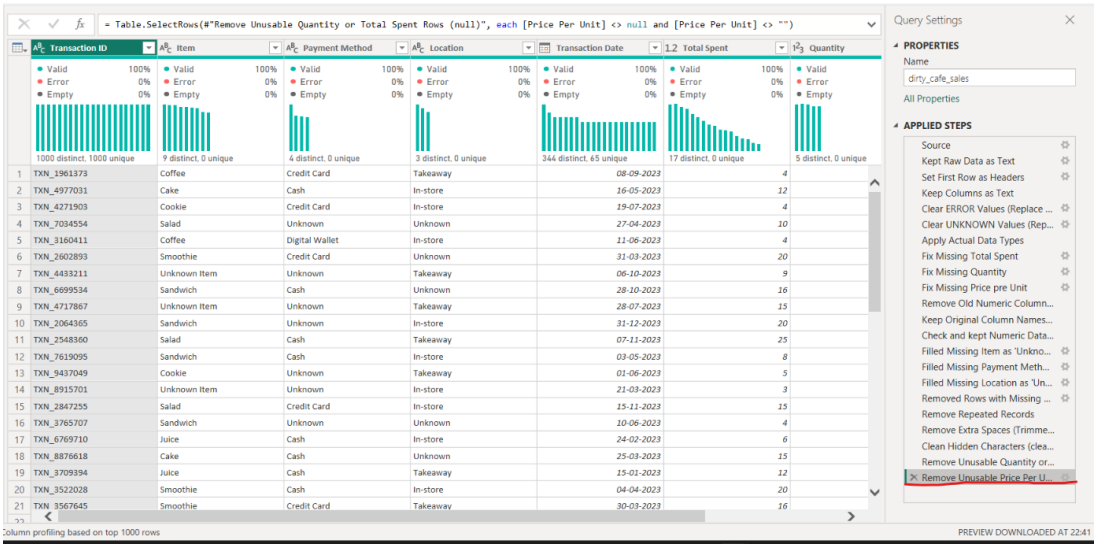
21. Filtered Rows1 → rename to " Remove Unusable Quantity or Total Spent Rows (null) ":

- Removed rows where Quantity or Total Spent were still blank after the calculation steps (i.e. both values were missing and could not be recovered).
- This only affected a small number of rows where more than one numeric field was blank at the same time, meaning the calculation logic in Steps 8–10 had nothing to work from.



22. Filtered Rows2 → rename to "Remove Unusable Price Per Unit Rows (null)":

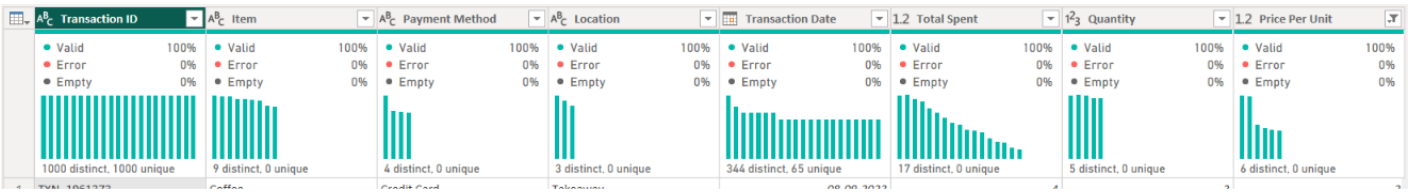
- Removed any remaining rows where Price Per Unit was still blank.
- This was the final safeguard step, catching any last rows where recovery genuinely wasn't possible before the dataset was considered clean.

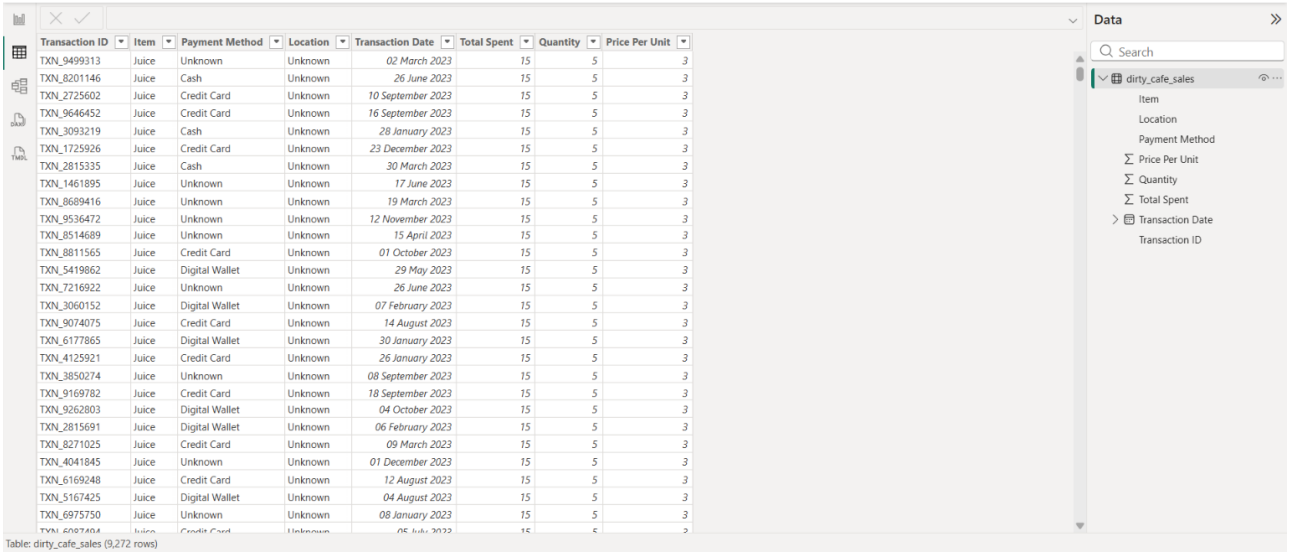


3. Output:

After all cleaning steps were applied, the dataset was loaded into the Power BI data model. The cleaned table shows the following results:

- Final row count: 9,272 (reduced from the original 10,000 rows).
- All 8 columns have the correct data type: Quantity (Whole Number), Price Per Unit and Total Spent (Decimal Number), Transaction Date (Date), and the remaining columns as Text.
- Zero blank or placeholder ("ERROR"/"UNKNOWN") values remain in any column.
- Zero duplicate records remain in the dataset.
- Missing Item, Payment Method, and Location values were preserved as explicit "Unknown" entries rather than deleted, since these columns had a high proportion of missing data and dropping them would have discarded a large share of the dataset.
- 728 rows (7.3% of the original dataset) were removed in total, limited strictly to rows with a missing transaction date, or rows where a numeric value could not be recovered through calculation.





Transaction ID	Item	Payment Method	Location	Transaction Date	Total Spent	Quantity	Price Per Unit
TXN_9499313	Juice	Unknown	Unknown	02 March 2023	15	5	3
TXN_8201146	Juice	Cash	Unknown	26 June 2023	15	5	3
TXN_2725602	Juice	Credit Card	Unknown	10 September 2023	15	5	3
TXN_9646452	Juice	Credit Card	Unknown	16 September 2023	15	5	3
TXN_3093219	Juice	Cash	Unknown	28 January 2023	15	5	3
TXN_1725926	Juice	Credit Card	Unknown	23 December 2023	15	5	3
TXN_2815335	Juice	Cash	Unknown	30 March 2023	15	5	3
TXN_1461895	Juice	Unknown	Unknown	17 June 2023	15	5	3
TXN_8689416	Juice	Unknown	Unknown	19 March 2023	15	5	3
TXN_9536472	Juice	Unknown	Unknown	12 November 2023	15	5	3
TXN_8514689	Juice	Unknown	Unknown	15 April 2023	15	5	3
TXN_8811565	Juice	Credit Card	Unknown	01 October 2023	15	5	3
TXN_5419862	Juice	Digital Wallet	Unknown	29 May 2023	15	5	3
TXN_7216922	Juice	Unknown	Unknown	26 June 2023	15	5	3
TXN_3060152	Juice	Digital Wallet	Unknown	07 February 2023	15	5	3
TXN_9074075	Juice	Credit Card	Unknown	14 August 2023	15	5	3
TXN_6177865	Juice	Digital Wallet	Unknown	30 January 2023	15	5	3
TXN_4125921	Juice	Credit Card	Unknown	26 January 2023	15	5	3
TXN_3850274	Juice	Unknown	Unknown	08 September 2023	15	5	3
TXN_9169782	Juice	Credit Card	Unknown	18 September 2023	15	5	3
TXN_9262803	Juice	Digital Wallet	Unknown	04 October 2023	15	5	3
TXN_2815691	Juice	Digital Wallet	Unknown	06 February 2023	15	5	3
TXN_8271025	Juice	Credit Card	Unknown	09 March 2023	15	5	3
TXN_4041845	Juice	Unknown	Unknown	01 December 2023	15	5	3
TXN_6169248	Juice	Credit Card	Unknown	12 August 2023	15	5	3
TXN_5167425	Juice	Digital Wallet	Unknown	04 August 2023	15	5	3
TXN_6975750	Juice	Unknown	Unknown	08 January 2023	15	5	3

4. Conclusion:

This experiment involved importing a raw, inconsistent cafe sales dataset into Power BI and cleaning it using Power Query. **Placeholder text, missing values, incorrect data types, and duplicate records were identified and systematically resolved.** Where possible, missing numeric values were recovered through calculation rather than deleted, preserving as much of the original data as possible, while rows with unrecoverable information were removed with clear justification. The result is a clean, consistent dataset of 9,272 records, ready for further analysis and reporting.

5. Learning Outcomes:

Through this experiment, I learned:

- How to import a CSV file into Power BI using Get Data and Power Query.
- How to identify data quality issues using Power Query's Column Quality and Column Profile tools.
- How to replace placeholder text values and convert them into proper null values.
- How to correctly assign data types to text, numeric, and date columns.
- How to use custom columns and conditional logic (if...then...else) to recover missing numeric values from related columns.
- How to identify and remove duplicate records based on a defined set of columns.
- How to clean and standardize text data using Trim and Clean transformations.
- The importance of renaming Applied Steps clearly, and of thinking carefully about the order of transformations, to keep a Power Query process readable, correct, and easy to review.